15

Autumn 2024 – DPS - MID-SEM

Level : 1	Programme : MS, MP, MR, IP, RS			
Sl. No.	Course Co	Course Name	Type	Instructor(s)
1	PH1101	Mechanics I	Theory	Golam Mortuza Hossain
Level : 2	Programme : MS, MP, MR, IP, RS			
Sl. No.	Course Co	Course Name	Type	Instructor(s)
1	PH2101	Waves and optics	Theory	Rangeet Bhattacharyya
2	PH2102	Mechanics II	Theory	Ananda Dasgupta
Level : 3	Programme : MS, MP, MR, IP, RS			
Sl. No.	Course Co	Course Name	Type	Instructor(s)
1	PH3101	Classical Mechanics	Theory	Rajesh Kumble Nayak
2	PH3102	Quantum Mechanics	Theory	Siddhartha Lal
3	PH3103	Mathematical Methods of Physics	Theory	Koushik Dutta
4	PH3104	Electrical Circuits and Electronics	Theory	Dhananjay Nandi
Level : 4	Programme : MS, MP, MR, IP, RS			
Sl. No.	Course Co	Course Name	Type	Instructor(s)
1	PH4101	Condensed Matter Physics	Theory	Subhasis Sinha
2	PH4102	Introductory Astrophysics	Theory	Sudip Kumar Garain
3	PH4104	Nonlinear Dynamics	Theory	Anandamohan Ghosh
4	PH4105	Advanced Mathematical Methods of Physics	Theory	Ananda Dasgupta
5	PH4106	Field Theory and Relativistic Quantum Mechanics	Theory	Ritesh Kumar Singh
6	PH4107	Advanced Electricity, Magnetism, and Optics	Theory	Nirmalya Ghosh
7	PH4110	Soft Condensed Matter Physics	Theory	Arindam Kundagrami
Level : 5	Programme : MS, MP, MR, IP, RS			
Sl. No.	Course Co	Course Name	Type	Instructor(s)
1	PH5103	Advanced Condensed Matter Physics	Theory	Amit Ghosal

Indian Institute of Science Education and Research Kolkata

Department of Physical Sciences

PH1101 :: Mechanics I
Mid-semester Examination (Autumn 2024)
Marks: 20. Duration: 90 minutes

1. Compute the following limit: [2]

$$\lim_{x \to \pi} \frac{\cos\left(\frac{x}{2}\right)}{x - \pi} \quad .$$

2. Using the definition of derivative, compute $\frac{d}{dx}\left(e^{3x}\right)$. [3]

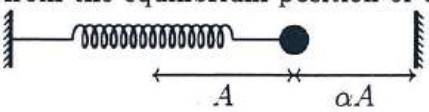
3. Draw two curves $f_1(x) = \sin(2x)$ and $f_2(x) = \cos(2x)$ in the domain $[0, \frac{\pi}{4}]$. Shade the region between these two curves in the given domain and then find the area of the shaded region. [2+3]

4. Find the locations and the corresponding values of the finite maxima and minima of the potential function $U(x) = x^4 - x^2$. [5]

5. An observer at a sea beach near the Earth's equator finds that the Sun takes about 2 min and 20 seconds to fully disappear below the horizon during the sunset. Estimate the radius of the Sun, in the unit of radius of the Earth r_E, if the distance to the Sun is 23450 r_E. [5]

G.M. Hossain

1. Consider a spring-mass oscillator of time period T. A wall is placed αA distance to the right from the equilibrium position of the oscillator, as shown below. [5]



The oscillator is given an initial displacement A towards the left and released from the rest. Consider all collisions to be elastic.

(a) Find the time it takes for the oscillator to move from the equilibrium to the wall. [2]

(b) Now find the time period of the oscillator in terms of α and T? [2]

(c) What is the period if $\alpha = \sqrt{3}/2$? [1]

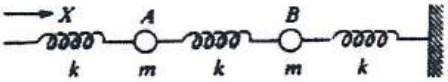
2. Consider two pendulums, a and b, with the same string length L, but with different bob masses, M and $3M$. They are coupled by a spring of spring constant K which is attached to the bobs. Assuming small angle oscillations, [5]

(a) Find the equations of motion using angles of the pendulums (w.r.t. the vertical) as dynamical variables. [2]

(b) Find the normal mode frequencies. [2]

(c) Find the ratios between normal mode amplitudes. [1]

3. Two equal masses m are connected to three identical springs (spring constant k) on a frictionless horizontal surface (see figure). One end of the system is fixed: the other is driven back and forth via a displacement $X = X_\circ \cos \omega t$. [5]



(a) Find the equations of motion of both the masses. [2]

(b) What are the normal frequencies? Hint: assume the left boundary to be fixed with $X_\circ = 0$ for this part of the problem. [2]

(c) What are the ratio of amplitudes of the masses when ω is equal to the lowest normal mode angular frequency? [1]

4. While moving, a mass m experiences a resistive force $-m\gamma v$ where v is the velocity, and γ is a constant, but no spring-like restoring force. [5]

(a) Show that its displacement as a function of time is of the form [2]

$$x = C - \frac{v_0}{\gamma} e^{-\gamma t},$$

where, C and v_0 are constants.

(b) At $t = 0$ the mass is at rest at $x = 0$. At this instant a driving force $F = F_0 \cos \omega t$ is switch on. Find the values of A and δ in the steady state solution $x = A \cos(\omega t - \delta)$. [1]

(c) Write down the general solution [the sum of parts (a) and (b)] and find the values of C and v_0 from the conditions that $x = 0$ and $\frac{dx}{dt} = 0$ at $t = 0$, and $\omega = \gamma$. [2]

Ranjeet Bhattacharya

PH2102 Mid Semester Examination

Full marks : 20 Time : **90 minutes**

Answer any two

Q 1) Bob is standing at the back of a train compartment which he observes to be 16 m long. The train is moving at a speed of $0.6c$ with respect to Alice, who is standing beside the track. He throws a ball at a speed of $0.8c$ towards the front wall of the carriage. The ball hits the front wall and rebounds perfectly elastically back to him. Consider the three events -

- $\mathcal{E}_1$: Bob throws the ball
- $\mathcal{E}_2$: The ball hits the front wall
- $\mathcal{E}_3$: The ball returns to Bob.

a) What is the temporal difference between $\mathcal{E}_2$ and $\mathcal{E}_1$ according to Alice?

b) What is the temporal difference between $\mathcal{E}_3$ and $\mathcal{E}_2$ according to Alice?

c) For which pair of events is one of the two measuring proper time?

d) Show that your results above is consistent with the time dilation formula (where it is appropriate).

For this problem use a lt-m= $\frac{1\,\mathrm{m}}{c}$ as the unit of time. *Hint : it may be a better idea to work with fractions (rather than use a calculator)* $(3+3+1+3)$

Q 2) a) Show how the length contraction formula can be derived using the K-calculus.

b) Derive an expression for the acceleration four-vector $A^\mu \equiv \frac{dU^\mu}{d\tau}$ in terms of velocity and acceleration 3 vectors. Is this four-vector time-like, light-like

Anindya Dey Gupta

or space-like? *Hint : the last part can be answered quickly if you take into account the fact that this classification is* observer independent.$(4 + (4 + 2))$

Q 3) A Lorentz transformation can be written in the compact form $\mathbf{x}' = L\mathbf{x}$ (where $\mathbf{x}$ is a 4×1 column vector with elements $x^0 = ct, x^1 = x, x^2 = y$ and $x^3 = z$). Let $\Lambda^{\mu}{}_{\nu}$ represent the μ-th row, ν-th column element of the 4×4 matrix L.

a) Use the fact that $\mathbf{x}^T \eta \mathbf{x}$ is an invariant, prove that $L^T \eta L = \eta$.

b) Write the equation in part (a) in terms of the components $\Lambda^{\mu}{}_{\nu}$ of the matrix L.

c) Show that $\det L$ must be either +1 or -1.

d) Show that $|\Lambda^0{}_0| \geq 1$.

e) It is given that the matrix L given below represents a particular Lorentz transformation:

$$\frac{1}{64}\begin{pmatrix} 125 & -48 & -60 & 75 \\ -75 & 80 & 36 & -45 \\ -60 & 0 & 80 & -36 \\ 48 & 0 & 0 & 80 \end{pmatrix}$$

Calculate L^{-1}. *Hint : it should take you no more than a minute or two, at most!* $(3 + 1 + 2 + 2 + 2)$

PH3101, Spring 2024

Mid-Term Exam

DPS, IISER Kolkata

Instructor: Dr. Rajesh Kumble Nayak
3:00 PM, 25 September, 2024.
Duration 90 Min.

- Read each question carefully.
- Answer all questions.
- Ensure all parts of the answer for a given question are in the same place.
- Write neatly, and do all rough work on the answer sheet.
- Do not write on the question paper.
- Keep answers concise and to the point.

Good luck!

Q - 1(20 Marks)
In a generalised coordinate system $\{\sigma, \tau\}$ the Lagrangian is given by,

$$\mathcal{L} = \frac{m}{2}\left(\frac{1}{\sigma^2+\tau^2}\right)\left[\dot{\sigma}^2+\dot{\tau}^2\right]$$

1. Find the Hamiltonian for the system
2. Find the Hamiltonian Equation of motion.

Q - 2(20 Marks)
In a generalised coordinate system $\{q_1, q_2\}$ the Lagrangian is given by,

$$\mathcal{L} = \dot{q}_1^2 + \dot{q}_2^2 + k_1 {q_1}^2 + k_2 \dot{q}_1 \dot{q}_2 ,$$

1. Find the generalised momenta $\{p_1, p_2\}$.
2. Express the generalised velocities $\{\dot{q}_1, \dot{q}_2\}$ in terms of generalised momenta $\{p_1, p_2\}$.
3. Find the Hamiltonian for the system. It is not necessary to simplify the Hamiltonian to its simplest form.

Q - 3(5 Marks)
For a Hamiltonian system, the Hamiltonian $H(q_i, p_i, t)$ satisfies the Hamiltonian equations of motion. Show that if $\frac{\partial H}{\partial t} = 0$, then $\frac{dH}{dt} = 0$ or H is constant of motion.

Q - 4(5 Marks)
Lagrangian for a system is given by

$$\mathcal{L} = \frac{m}{2}\left(\dot{x}^2+\dot{y}^2\right) + q\left(\dot{x}A_x + \dot{y}A_y\right) - q\Phi(x,y)$$

Here, $\vec{A} = (A_x, A_y)$. Find the Hamiltonian.

IISER–Kolkata
Classical Mechanics
Mid-semester supplementary examination

25/09/2024, 3 PM to 4.30 PM

Answer question 1 and any two from the rest.

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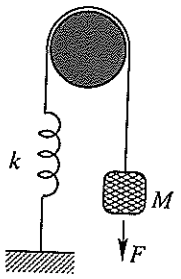
1. (a) Why does the moment of inertia of a rigid body have to be represented by a tensor? 2

   (b) What is a virtual displacement? 2

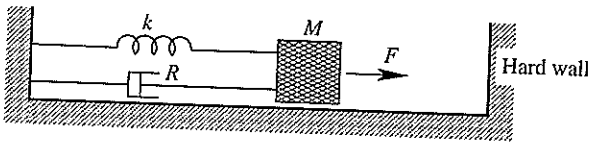
   (c) What is the 'Least Action Principle'? 2

   (d) Show that for a Hamiltonian system, $\dot{q}_j = \{q_j, H\}$ and $\dot{p}_j = \{p_j, H\}$ 4

2. Define 'holonomic constraint'. Are the following systems holonomic or non-holonomic? Write down the constraint equations (or inequations) for each system. (7+8)

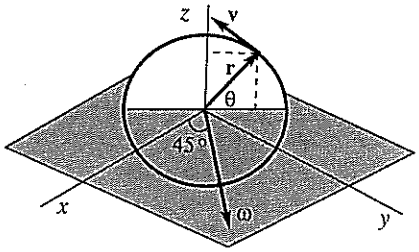


(a)



(b)

3. A particle is rotating in a vertical plane. Its angular velocity vector lies in the xy plane and makes an angle of 45° with the x and y axes.



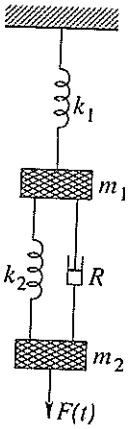
   (a) Find the vector $\vec{r}$ (3)

   (b) Find the vector $\dot{\vec{r}}$ by differentiating $\vec{r}$ (3)

   (c) Find the cross product $\vec{\omega} \times \vec{r}$ (3)

   (d) Find the angular momentum of the particle. (6)

4. Obtain the Lagrangian equations of motion for the system (15)



Siddhartha Lal
28/9/24

PH3102 Quantum Mechanics **Mid-semester Examination**

Instructor: Dr. Siddhartha Lal **Autumn Semester, 2024**

Instructions: Answer all questions for full marks. All necessary formulae are given. Hand in question paper together with your answer script. Put your name and roll no. on both Question paper as well as the Answer script.

Date: September 28, 2024 **Full marks: 20** **Time: 1.5 hours**

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1 Angular Momentum Eigenstates. [6 marks]

Given two angular momentum $\vec{J}_1$ and $\vec{J}_2$ such that $j_1 = 1$ and $j_2 = 1/2$ (where j_α $(\alpha = 1, 2)$ denotes the eigenvalue of the J_α^2 operator as $j_\alpha(j_\alpha + 1)\hbar^2$), compute the eigenstates of $\vec{J}_{tot} = \vec{J}_1 + \vec{J}_2$ (i) for the sector $j_{tot} = 3/2$, and (ii) for the sector $j_{tot} = 1/2$. Show all calculations explicitly.
Hint: You may find the following relation useful:

$$J_\pm \left|j, m_j\right\rangle \;=\; \hbar\sqrt{j(j+1) - m_j(m_j \pm 1)}\,\left|j, m_j \pm 1\right\rangle \;.$$

2 Larmor precession in the Heisenberg picture. [4 marks]

Consider the Hamiltonian H of a particle of spin $\vec{S}$ placed in a magnetic field: $H = BS_z$, where S_z is the spin operator in the $z-$direction. By starting from the respective Heisenberg equations of motion, calculate the rate of change dS_i/dt $(i \in (x, y, z))$ for all three spin operators S_x, S_y and S_z. Given that the initial values of these spin operators at time $t = 0$ are $S_i(0)$ $(i \in (x, y, z))$, solve all three equations to obtain the spin operators $S_i(t)$ as a function of time.

3 The quantum virial theorem. [6 marks]

For a one-dimensional system described by the Hamiltonian $H = \frac{p^2}{2m} + a|x|^n$, where $a > 0$ and $n > -2$, the Virial theorem states that in any eigenstate of H, the following relations are satisfied by the kinetic energy operator $\hat{T}$ and potential energy operator $\hat{V}$

$$\langle\hat{T}\rangle = \langle\frac{p^2}{2m}\rangle = \frac{n}{n+2}\langle H\rangle \;\;,\;\; \langle\hat{V}\rangle = \langle a|x|^n\rangle = \frac{2}{n+2}\langle H\rangle \;\;,\;\; 2\langle\hat{T}\rangle = n\langle\hat{V}\rangle \;. \tag{1}$$

(a) Consider the Hermitian operator $\hat{D} = [\hat{x}, \hat{p}_x]_+ /2\hbar = (\hat{x}\hat{p}_x + \hat{p}_x\hat{x})/2\hbar$ (i.e., $[\,]_+ \equiv$ the anti-commutator of $\hat{x}$ and $\hat{p}_x$) and compute the commutator $\left[\hat{D}, H\right]$.

(b) Prove that for any Hermitian operator $\hat{\Theta}$, the following expectation value computed using the stationary eigenstates ($|\psi\rangle$) of the Hamiltonian (H) must vanish

$$\langle\psi|\left[\hat{\Theta}, H\right]|\psi\rangle = 0 \;. \tag{2}$$

(c) Use the results of parts (a) and (b) to prove the virial theorem.
(d) What is the outcome from the theorem for (i) the case of $n = 2$ (the simple harmonic oscillator) and (ii) the case of Coulomb inter-particle interactions with $n = -1$?

4 The Hellman-Feynman theorem. [4 marks]

For a Hamiltonian H that is a function of a parameter λ, $H \equiv H(\lambda)$, and an eigenstate $|\psi_n\rangle$ with eigenvalue E_n, the Hellmann-Feynman theorem states

$$\frac{\partial E_n}{\partial \lambda} = \langle \psi_n | \frac{\partial H(\lambda)}{\partial \lambda} | \psi_n \rangle \ . \tag{3}$$

Recall that the harmonic oscillator Hamiltonian $H = \frac{p^2}{2m} + \frac{m\omega^2 x^2}{2}$ is dependent on two parameters, m (mass) and ω (frequency). Use the eigenvalues of the simple harmonic oscialltor ($E_n(\omega)$) and choose the parameter λ carefully so as to implement the Hellmann-Feynman theorem and prove the following for the harmonic oscillator problem:

(a) $\langle \frac{p^2}{2m} \rangle = \langle \frac{m\omega^2 x^2}{2} \rangle$, (b) $\langle \frac{m\omega^2 x^2}{2} \rangle = \frac{1}{2}(n + 1/2)\hbar\omega$.

Siddhartha Lal
28/9/24

PH3103 Mathematical Methods of Physics
Autumn Semester - 2024
Indian Institute of Science Education and Research, Kolkata
Instructor: Koushik Dutta

Mid-Semester Examination Examination Date: 27/09/2024

1. Starting from the generating function $G(x,t)$ for the Bessel function, establish the recurrence relation $2J_n^{'} = J_{n-1} - J_{n+1}$ where this proof is restricted to positive integer n. In the class, we have derived this recurrence relation for general ν. It is given that $G(x,t) = exp\left[\frac{x}{2}(t - 1/t)\right]$.
Marks: 5

2. Consider $f(x) = sin\ x$ and $g(x) = 1 - sin\ x$. Find out whether $f(x)$ and $g(x)$ are linearly dependent or independent. **Marks: 3**

3. *Note: Each part of this problem is independent. So, you may wish to proceed to the next part, if you are stuck somewhere!*

Consider the following time-independent Shrodinger equation in one dimension

$$\left(-\frac{\hbar^2}{2m}\frac{d^2}{dx^2} + V(x)\right)\psi(x) = E\psi \tag{1}$$

for a quantum mechanical particle of mass m under the simple harmonic potential $V(x) = \frac{1}{2}m\omega^2x^2$. Here E is the energy eignevalues of the particle. The goal is to solve the above equation to find out energy of the particle.

(a) Make the following change of variables $\xi = \sqrt{\frac{m\omega}{\hbar}}x$, $\epsilon = \frac{2E}{\hbar\omega}$ to show that the above equation can be written as

$$\frac{d^2\psi(\xi)}{d\xi^2} + (\epsilon - \xi^2)\psi(\xi) = 0. \tag{2}$$

We have implicitly assumed $\psi(x) = \psi(\xi)$, which is technically not correct! But, let us not worry about it. **Marks: 2**

(b) For finite values of ξ, does the Eq. (2) has any singularity? Find out the nature of singularity of Eq. (2) at $\xi = \pm\infty$. **Marks: 1+4**

(c) Do a series solution of Eq. (2) around $\xi = 0$, and find out the recurrence relation connecting several coefficients of the expansion. How many terms are there in the recurrence relation? Write $\psi(\xi)$ in terms of two arbitrary constants upto order ξ^4. **Marks: 6**

(d) We try the following trial solution

$$\psi(\xi) = f(\xi)e^{-\frac{\xi^2}{2}} \tag{3}$$

for Eq. (2), where $f(\xi)$ is a polynomial of positive powers only. Show that $f(\xi)$ satisfies the following differential equation:

$$\frac{d^2f(\xi)}{d\xi^2} - 2\xi\frac{df(\xi)}{d\xi} + (\epsilon - 1)f(\xi) = 0. \tag{4}$$

Marks: 4

(f) Do a series solution of Eq. (4) around $\xi = 0$ to write $f(\xi)$ in terms of two unknown coefficients. How many terms are there in this recurrence relation as compared to what you found in part (c)? **Marks: 5**

(h) To make the $\psi(\xi)$ normalisable, we forcefully 'chop-off' the series to make it a polynomial of finite order, and show that it leads to the usual formula of energy quantisation

$$E_n = \left(n + \frac{1}{2}\right)\hbar\omega, \quad n = 0, 1, 2, \ldots \tag{5}$$

Marks: 2

(i) Write down the expression for $\psi_0(x)$ and $\psi_1(x)$ and plot these functions. Do not worry about the overall constant factors. **Marks: 2**

4. The ground state wave function for an one dimensional simple harmonic oscillator is given by

$$\psi_0(x) = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} exp\left(-\frac{m\omega}{2\hbar}x^2\right) \tag{6}$$

Doing the Fourier transformation, find the momentum space wave function $\tilde{\psi}_0(k)$. Find $\Delta x \Delta p$ for the ground state, where $\Delta A = \sqrt{\langle A^2\rangle - \langle A\rangle^2}$, and $\langle A\rangle = \int_{-\infty}^{\infty} A|\psi(x)|^2 dx$.
It is given $\int_0^\infty \xi^2 exp(-a\xi^2) = \frac{\sqrt{\pi}}{4}a^{-3/2}$; $Re(a) > 0$. **Marks: 5+5**

5. Prove the following identity

$$e^{i\vec{k}\cdot\vec{r}} = 4\pi \sum_{l=0}^{\infty} \sum_{m=-l}^{l} (i)^l j_l(kr) Y_l^m(\Omega_k) Y_l^{-m}(\Omega_r) \tag{7}$$

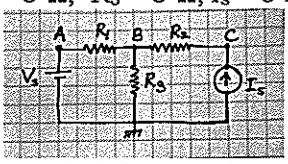
It is given $j_l(kr) = \frac{1}{2i^l}\int_{-1}^{1} e^{ikrt} P_l(t) dt$, and use addition theorem of spherical harmonics. Here $j_l(kr)$ are spherical Bessel function. **Marks: 6**

04/10/2024

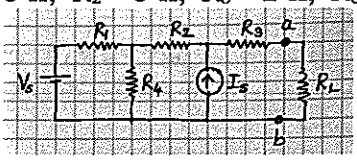
Course: PH3104 : Electrical Circuits & Electronics
Mid-Sem Examination (Autumn 2024)

Full Marks: 20 Time: 1.5 Hrs

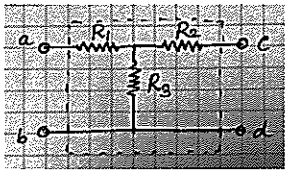
1. (a) Using **Node Voltage Method (NVM)**, calculate the potential at points A, B and C. Consider: $V_s = 70$ V, $R_1 = 10\ \Omega$, $R_2 = 6\ \Omega$, $R_3 = 5\ \Omega$, $I_s = 5$ A [3]



(b) Find the value of Load Resistance (R_L) for **maximum power transfer** in the following circuit. Consider $V_s = 12$ V, $R_1 = 6\ \Omega$, $R_2 = 3\ \Omega$, $R_3 = 2\ \Omega$, $R_3 = 12\ \Omega$, $I_s = 2$ A [2]

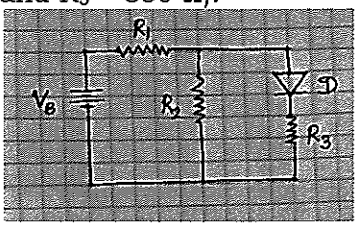


2. (a) When is a two-port network said to be **reciprocal**? Write the condition using z-parameter for which the network is reciprocal. [1]
(b) For the given two-port network, determine the **h-parameter matrix**. (Consider: $R_1 = R_2 = R_3 = 10\ \Omega$). [4]



3. A 30 Ω resistor, a 20 mH inductor and a 2 μF capacitor are all connected in **series** across a 9 V, varying frequency supply. Calculate:
(a) Resonant frequency.
(b) Circuit current.
(c) Inductive reactance.
(d) Voltage across the inductor and the capacitor.
(e) Quality Factor and Bandwidth. [5]

4. (a) What is the junction capacitance of a **p-n junction diode** and how does it form? [1]
(b) Draw **static and dynamic** characteristics of a p-n junction diode. Label the diagram clearly and mark the **operating point**. Explain with diagram, how the operating point changes if the input voltage **(V_i)** and load resistance **(R_L)** changes independently keeping other as constant. [2]
(c) The p-n junction diode used in the following circuit has a ***cutin*** voltage of 0.6 V and a forward resistance of 150 Ω. If the diode can dissipate a maximum power of 200 mW, calculate the **maximum permissible value** of the battery voltage V_B.
(Consider: $R_1 = 6$ kΩ, $R_2 = 3$ kΩ and $R_3 = 850\ \Omega$). [2]



Condensed Matter Physics(PH4101) Mid Sem Exam

time 1.5hr

1. Consider 2-dimensional electron gas with density n (N/area), at $T = 0$.

a) Find out: i). Density of states, ii)Fermi momentum and Fermi energy, iii) Energy per particle in terms of Fermi energy, iv) Kinetic energy density in terms of local density n. (2+2+2+2)

b)Consider this system in weak magnetic field B and an additional Zeeman term in the Hamiltonian $\mu_B B \sigma_z$. Find out magnetisation and susceptibility. (5).

2. Consider 3-dimensional solid with phonon dispersion $\omega = v|k|$, which is valid upto Debye cutoff ω_D.

a) Find out the density of states of phonon gas.

b) Using (a), write down energy of phonon gas E at temperature T. Obtain E at high temperature (keep leading term and next correction).

c) From (b), find out ω_D so that Dulong-Petit law remains valid. (2+4+1).

MidSem Exam - Autumn 2024
PH4102: Introductory Astrophysics

Answer Question 1 and any 4 out of the 5 remaining questions.

Note: Meaning of variables and symbols are same as we used in the class, unless stated otherwise. Clearly mention any assumption that you make while solving any problem.

Time : 90 min Marks: 30

(Given : $m_e = 9.1 \times 10^{-31}$ kg, $m_H = 1.67 \times 10^{-27}$ kg, $h = 6.62 \times 10^{-34}$ m^2 kg/s, $k_B = 1.38 \times 10^{-23}$ J/K, $G = 6.67 \times 10^{-11}$ m^3 kg^{-1} s^{-2}, 1 AU $= 1.5 \times 10^8$ km, $M_\odot = 1.99 \times 10^{30}$ kg, $R_\odot = 7 \times 10^8$ m, $L_\odot = 3.85 \times 10^26$ W)

1. a) Draw the H-R diagram and mark the positions of main sequence, giants, supergiants and white dwarfs on this. (2)

b) In general, a star is made up of hydrogen, helium and metals having mass fractions X, Y and Z, respectively. Derive an expression for the mean molecular weight μ that can be used to calculate the pressure for such a star. (2)

c) Derive an expression for Eddington luminosity. (2)

d) Using the definition of parsec, find its value in meter. (2)

e) Radiant flux is expressed as radiation energy passing through unit area per unit time. Find out at what distance from a 100 W light source (e.g., LED source) is the radiant flux equal to the Solar radiant flux at Earth? (2)

2. a) Write down the radiative transfer equation and meaning of each variable.

b) Derive the formal solution of this equation and discuss the physical significance of each term of the solution. (2 + 3)

3. a) Derive the equation of hydrostatic equilibrium. Clearly state all your assumptions.

b) Using this equation, make an order of magnitude estimate of pressure at the center of the Sun. (3+2)

4. In a fully degenerate gas, the relation between the electron number density n and the Fermi momentum P_F is

$$n = \frac{8\pi}{3h^3} P_F^3$$

a) In the non-relativistic gas, use the relation $P_F = \sqrt{2m_e E_F}$, where E_F is Fermi energy and express E_F in terms of n and m_e.
b) Assuming the a typical white dwarf mass 1 $M_\odot$ and radius 10000 KM, estimate n (assume oxygen/carbon core).
c) Using this n and $T = 10^7$ K, calculate the ratio of E_{th}/E_F, where E_{th} is the average thermal energy of an electron in a gas of temperature T.
d) Provide a physical interpretation of the value of this ratio. (1+1.5+1.5+1)

5. Briefly describe the evolution of a 5 $M_\odot$ star since its beginning of lifetime on the main sequence. Draw associated evolutionary track on the H-R diagram. (5)

6. a) Virial theorem for stars states that the total energy (i.e., sum of thermal and gravitational energy) is half of the potential energy. Prove this theorem starting from the equation of hydrostatic equilibrium. Mention clearly any assumption that you make.
b) In the early years of stellar research, it was believed that part of the gravitational energy release due to contraction was the source of observed luminosity of the star. Show that such assumption leads to a shorter lifetime of the Sun. (3+2)

Non-Linear Dynamics PH4104

Mid-Semester Examination 25 September 2024

Duration: 1.5 hours. Full Marks: 4 X 5 = 20 Attempt all questions

Problem 1

Determine the conditions for Lyapunov stability of the system:

$$\begin{aligned} \frac{dx}{dt} &= -x\,y^2 \\ \frac{dy}{dt} &= -a\,y - y\,x^2 \end{aligned}$$

Problem 2

Plot the bifurcation diagram of the map $x_{n+1} = x_n + r - x_n^2$, for the parameter $r \in [-1 : 1 + \epsilon]$.

Show the bifurcation points and determine the types of bifurcations.

Problem 3

Calculate the fixed points and determine their stability for the dynamical system

$$\begin{aligned} \frac{dx}{dt} &= -y - x^2 \\ \frac{dy}{dt} &= x - x^3 \end{aligned}$$

Draw the phase portrait showing the stable/unstable manifolds with arrows.

Problem 4

Find the conditions for which the enzyme reaction kinetics given by

$$\begin{aligned} \frac{du}{dt} &= a - u + u^2\,v \\ \frac{dv}{dt} &= b - u^2\,v \end{aligned}$$

show periodic oscillations.

PH4105 Mid Semester Examination

Total marks : 20 Time 90 mins

*Attempt any **two** questions*

Q 1) a) Show that a map X that acts on smooth real-valued functions on a manifold M to produce smooth functions that obey the properties

$$X(af+bg) = aXf + bXg$$
$$X(fg) = fXg + gXf$$

for all $f, g \in C^\infty(M)$ and $a, b \in \mathbb{R}$ defines a vector field on M.

b) Prove that the Lie bracket $[X, Y]$ of two vector fields X and Y, defined by

$$[X,Y]f = X(Yf) - Y(Xf), \qquad f \in C^\infty(M)$$

is a vector field.

c) Show that if (U, ϕ) is a chart on M with coordinate functions $(x^1, \ldots, x^n)$ then $\left[\frac{\partial}{\partial x^i}, \frac{\partial}{\partial x^j}\right] = 0$. $(3+4+3)$

Q 2) a) Consider the set $M = SL(2,\mathbb{R}) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} : (a,b,c,d) \in \mathbb{R}^4,\ ad - bc = 1 \right\}$. Show that the pair (U, ϕ) where

$$U = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL(2,\mathbb{R}) : a \neq 0 \right\}$$

and

$$\phi : \begin{pmatrix} a & b \\ c & d \end{pmatrix} \mapsto (a,b,c)$$

is a valid chart on M.

b) Show that the pair (V, χ) defined by

$$V = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL(2,\mathbb{R}) : d \neq 0 \right\}$$

and

$$\chi : \begin{pmatrix} a & b \\ c & d \end{pmatrix} \mapsto (b,c,d)$$

is another chart, which is $C^\infty(M)$ compatible with the chart (U, ϕ).

c) Do the two charts (U, ϕ) and (V, χ) form a C^∞-atlas on M? Explain. $(4+4+2)$

Q 3)a) Consider the one form

$$\alpha = \frac{xdy - ydx}{x^2 + y^2}$$

on $M = \mathbb{R}^2 \setminus \{(0,0)\}$ where (x, y) are the standard coordinate functions of the chart (M, id).

Determine $\int_C \alpha$ where the curve $C : [0, 2\pi] \to \cancel{\mathbb{R}^2}$ M is given by $t \mapsto (2\cos t, 3\sin t)$

b) Consider a $\binom{0}{2}$ tensor field on the manifold $\mathbb{R}^2$, given in the standard coordinate system by

$$t = y^2\, dx \otimes dx - x^2\, dx \otimes dy + y^2\, dy \otimes dx + xy\, dy \otimes dy$$

Express this field in the coordinate basis provided by polar coordinates (r, θ).

c) Let r_p, s_p and t_p be $\begin{pmatrix} 0 \\ k \end{pmatrix}, \begin{pmatrix} 0 \\ l \end{pmatrix}$ and $\begin{pmatrix} 0 \\ m \end{pmatrix}$ tensors at $p \in M$. Prove that

$$r_p \otimes (s_p \otimes t_p) = (r_p \otimes s_p) \otimes t_p$$

$(4+3+3)$

Date : 4th Oct, 2024 Mid-semester Exam

PH4106 : Quantum Field Theory

Duration: 10h00 to 18h00

Roll No: Name:

1. **Classical Yukawa Theory:** Consider the classical theory of a real scalar field ϕ and a Dirac field ψ given as

$$\mathcal{L}_1 = \frac{1}{2}(\partial_\mu\phi)(\partial^\mu\phi) - \frac{m^2}{2}\phi^2 - \frac{\lambda}{24}\phi^4 + \bar{\psi}(i\gamma^\mu\partial_\mu - M)\psi - g\bar{\psi}\psi\phi$$

   (a) Obtain Euler-Lagrange equation of motion for the fields.

   (b) Obtain the expression for the stress-energy tensor $T^{\mu\nu}$.

   (c) Compute conservation equation $\partial_\nu T^{\mu\nu} = 0$ and comment on what additional information does it contain, if any.

2. **Classical Yukawa Theory:** Consider the two real fields ϕ_1 and ϕ_2 satisfying following equations of motion:

$$a\nabla^2\phi_1 + b\frac{\partial\phi_1}{\partial t} - c\frac{\partial^2\phi_1}{\partial t^2} = f\phi_2$$
$$a\nabla^2\phi_2 - b\frac{\partial\phi_2}{\partial t} - c\frac{\partial^2\phi_2}{\partial t^2} = g\phi_1$$

   (a) Write the simplest Lagrangian density that gives the above set of Euler Lagrange equations of motion.

   (b) Obtain the stress-energy tensor $T^{\mu\nu}$ for the above Lagrangian density.

3. **Scalar QED and Classical Mechanics:** Consider a complex field Φ with electric charge e having *local phase invariance* or *the gauge invariance.*

   (a) Write down the Lagrangian density for the scalar and the gauge bosons. Expand the co-variant derivative and write all the terms explicitly.

   (b) Compute the cannonically conjugate momemtum for each field in the Lagrangian. Comment on the field momentum for A^0 component of the gauge field.

   (c) Consider above Lagrangian in $1+1$ dimension. Write the components of $F^{\mu\nu}$ tensor. Comment of electric and magnetic fields in $1+1$ dimensions.

   (d) For $1+1$ dimension, write down the spring-mass-ball model for the EM part ($F^{\mu\nu}$) of the Lagrangian and comment of the field A^0.

   (e) For $1+1$ dimensin, write down the spring-mass-ball model for the full Lagrangian.

4. **Dirac equation in $1+1$ dimension:**

   (a) Setup the Dirac equation in $1+1$ dimensions following the steps followed in the class, i.e. in terms of α and β matrices. Rewrite it in terms of γ matrices.

   (b) List the basic properties of α, β and γ matrices obtained above. Make a choice for γ matrices satisfying the properties in $1+1$ dimension.

(c) Find the spinor solution for the above choice of γ matrices and discuss the normalizations.

(d) Obtain the energy projection operator.

(e) For the massless case, show that the spinors are eigen values of chirality operator (suitably defined version of γ^5).

(f) Construct the Clifford algebra in $1+1$ dimension and argue the linear independence.

(g) Obtain the complete set of *four spinor* Fiertz identities starting from the basic *two spinor* Fiertz identity in $1+1$ dimension.

(h) Obtain the Gordon like Idendity in $1+1$ dimension and comment of peculiarities you face.

(i) Upgrade above Dirac equation to include static background field. Solve it for a step potential ($V(x) = 0$ for $x < 0, V(x) = V$ for$x \geq 0$). Consider the solutions for energy $m < E < V$. Find the condition on V for which the solution in $x > 0$ region is non-decaying.

IISER Kolkata
Department of Physical Sciences
Advanced Electricity, Magnetism and Optics, PH4107
Autumn Semester 2024
Mid semester Examination, dated 30/09/2024
Max. Marks: 30

Q1. 4+3+3

(a) Draw typical dispersion (dependence on frequency ω) curves for the real and imaginary parts of refractive index of a medium which would lead to – (i) group velocity of light $v_g \gg c$, and (ii) $v_g \ll c$. What happens to the phase velocity v_p in such scenario? Does (i) imply that signal can propagate with a speed greater than c? Explain your answer briefly.

(b) What is the origin of the negative value of the real part of dielectric permittivity $\varepsilon'(\omega)$ in metals? What happens to the electric and magnetic fields of light when it encounters a medium having $\varepsilon'(\omega) = 0$? Can such behavior be observed for dielectric medium also?

(c) Comment when would 'plasmon' be a well-defined entity among the following two cases –
(i) $\frac{Ne^2}{m\varepsilon_b} = 16\times10^{32}\ s^{-2}$, $\gamma = 5\times10^{13}\ s^{-1}$ and (ii) $\frac{Ne^2}{m\varepsilon_b} = 6\times10^{31}\ s^{-2}$, $\gamma = 10^{17}\ s^{-1}$

What happens to the relative ratio of time averaged electric and magnetic field energy densities when $\omega = \sqrt{\frac{Ne^2}{m\varepsilon_b}} \gg \gamma$. What would be the magnitude of the Poynting vector S? All the symbols carry usual meaning.

Q2. 6+4

(a) The electric polarization of some material responds to an electric field step ($E(t)$) in the following way $P(t) = \chi_1 E_0(1 - t/\tau)$*, where $E(t) = 0$ for $t < 0$ and $E(t) = E_0$ for $t > 0$. Here χ_1 and $\tau > 0$ are some constants. Determine frequency dependent susceptibility $\chi(\omega)$ of this material.

(b) How would the electric polarization $P(t)$ vary for an input δ-functional pulse electric field $E(t)$?

Q3. 3+4+3

In a dispersive medium, the frequency dependent current density and the electric field are related as $\boldsymbol{J}(\omega) = \sigma(\omega)\,\boldsymbol{E}(\omega)$.

(a) Show that $\boldsymbol{J}$ and $\boldsymbol{E}$ have non-local connection in time.

(b) Determine the poles of the frequency dependent conductivity $\sigma(\omega)$ and show that the non-local temporal connection between $\boldsymbol{J}$ and $\boldsymbol{E}$ obeys the principle of causality.

(c) Derive the temporal response function and comment on its nature. What is the characteristic time scale of non-locality?

* $P(t) = E_0[1 - \exp(-t/\tau)]$

Midsem Exam: PH4110, SOFT CONDENSED MATTER PHYSICS, AUTUMN 2024

MARKS: 20, TIME: 1:30 HOURS, 01 October 2024, 3:00 PM

ANSWER ALL QUESTIONS. MARKS WILL BE SCALED TO FULFIL THE MARKING SCHEME. IF RUNNING SHORT IN TIME, ANSWER AS MUCH AS YOU CAN. WRITE EVERY STEP OF YOUR CALCULATION. GOOD LUCK!

1. (i) Draw the $p - n$ and $p - T$ phase diagrams of water. Identify the critical point in terms of all thermodynamic variables. Illustrate the path of the system under constant pressure but increasing temperature (both for temperature ranges below T_c and above T_c) and identify first and second order transitions.

(ii) Show why, even when the functional dependence of pressure in a chemical system is $p = p(n, T)$, the vapour pressure of a liquid only depends on temperature but not on temperature and density independently.

[4 marks]

2. (i) Show that, in a fluid, $dg = -sdT + vdp$, where g is the Gibbs free energy per particle, s is the entropy per particle, and v is the specific volume.

(ii) Explain with illustrations why a liquid is more correlated than a gas.

[4 marks]

3. Consider a periodic solid. If $\langle x^2 \rangle$ is the measure of thermal fluctuation of displacement where x is the displacement of an atom from its equilibrium position, show that the fluctuation is proportional to the temperature and inversely proportional to the rigidity modulus.

[4 marks]

4. (i) How much heat must be added to a large system at 298K for the number of accessible states to increase by a factor of 10^6 ?

(ii) Calculate the energy of a system of N particles with spin one on each site, given that the levels associated with the three spin states have energies ϵ, 0, $-\epsilon$.

[4 marks]

Arindam Kundagrami
1/10/24

5. Explain, with sketches, how many different types of ordered phases a solution of disk-like particles can have. Find as many as you can.

[4 marks]

Formulae you may find useful (but may not require to use all of them):

1. The first law of thermodynamics:

$$dE = TdS - PdV + \mu dN$$

2.

$$C_v = \left(\frac{\partial E}{\partial T}\right)_{N,V}, C_p = \left(\frac{\partial (E+PV)}{\partial T}\right)_{N,P}$$

3.

$$F = E - TS, \qquad G = F + PV,$$

where F, G are Helmholtz and Gibbs free energies respectively.

4.

$$\left(\frac{\partial x}{\partial y}\right)_z \left(\frac{\partial y}{\partial z}\right)_x \left(\frac{\partial z}{\partial x}\right)_y = -1$$

5.

$$\Gamma(\nu) = \int_0^\infty e^{-x} x^{\nu-1} dx; \qquad \nu > 0$$

$$\Gamma(n) = (n-1)!, \qquad \Gamma(n+1/2) = (n-1/2)(n-3/2)...3/2.1/2.\sqrt{\pi}, \qquad \Gamma(1/2) = \sqrt{\pi}$$

6. Hyper-volume and hyper-surface area of n dimensional sphere of radius R:

$$V_n(R) = \frac{\pi^{n/2}}{(n/2)!} R^n, \qquad S_n(R) = \frac{2\pi^{n/2}}{\Gamma(n/2)} R^{n-1}$$

8. Stirling's formula:

$$\ln(N!) \simeq N \ln N - N, \qquad N \gg 1$$

9. 1 eV $= 1.6 \times 10^{-19}$ J, $h = 6.62 \times 10^{-34}$m^2.kg/s, $c = 3 \times 10^8$m/s, $m_e = 9.11 \times 10^{-31}$kg, $m_p = 1.67 \times 10^{-27}$kg, $k_B = 1.38 \times 10^{-23}$ m^2 kg s^{-2} K^{-1}.

17. Mean thermal wavelength, $\lambda = h/(2\pi m k_B T)^{1/2}$.

18. The grand potential in GCE is given by: $d\tilde{A} = -SdT - Nd\mu - PdV$.

19. $\ln(1+x) = x - x^2/2 + x^3/3 - x^4/4 + ..$

21. Single particle (non-relativistic classical gas) energy levels:

1d: $\epsilon = (h^2/8mL^2)n_x^2$

Arindam Kundagrami
1/10/24

PH5103 (Adv. CMP) Mid-semester exam, 25 September, 2024

1. (a) The current-density operator for one-particle in symmetried first quantized form is given by,

$$\hat{\vec{j}}(\vec{r}) = \frac{1}{2m}\left[\hat{\vec{p}}\delta(\hat{\vec{r}}-\vec{r}) + \delta(\hat{\vec{r}}-\vec{r})\hat{\vec{p}}\right].$$

Consider a system of free (spinless) many particles, so that the 1st quantized form of the Hamiltonian is:

$$\mathcal{H}_{\text{free}} = \sum_i \frac{\hat{p}_i^2}{2m}.$$

Express the current density operator in momentum-space, i.e. $\hat{\vec{j}}(\vec{q})$ in second quantized form in terms of creation and annihilation operators in momentum basis. Assuming that the eigenstates of the free Hamiltonian respects the underlying symmetries of the hamiltonian, derive the average current (at a temperature T) in the system described by the free Hamiltonian.
Note that, in second quantization (using class lectures notations) the N-particle current density operator is:

$$\hat{\vec{j}}(\vec{q}) = \sum_{k_1,k_2} \vec{j}^{(1)}_{k_1,k_2}(\vec{q})\hat{c}^\dagger_{k_1}\hat{c}_{k_2}, \quad \text{where,} \quad \vec{j}^{(1)}_{k_1,k_2}(\vec{q}) = \langle k_1|\hat{\vec{j}}^{(1)}(\vec{q})|k_2\rangle.$$

[points: 10]

2. (a) Obtain Pauli-spin susceptibility $\chi^{(0)}$ of a free electron gas. Pauli paramagnetism arises when electronic spins get coupled to the Zeeman magnetic field B causing additional Zeeman energy $\hat{\mathcal{H}}_Z = -\mu_B B \sum_{\vec{k},s} \sigma_z \hat{n}_{\vec{k}s}$. So, the energy of an electron is either raised or lowered depending on its spin. This causes Pauli paramagnetism of free electrons [Note, magnetization $M = \mu_B(n_\uparrow - n_\downarrow)$]. Your result of $\chi^{(0)}$ should involve the DOS at Fermi level of the free electron gas, $N^{(0)}(\varepsilon_F)$ and the components of $\mathcal{H}_Z$ already introduced.

(b) Noting that the defining equation for Landau's description of Fermi liquids (notations carry the meaning discussed in the class),

$$E = E_0 + \sum_{k,\sigma} \varepsilon_{k,\sigma}\delta n_{k,\sigma} + \frac{1}{2}\sum_{k\sigma,k'\sigma} f_{k\sigma,k'\sigma}\delta n_{k,\sigma}\delta n_{k',\sigma'}$$

and, $f_{k\uparrow,k'\uparrow} = f_{k\downarrow,k'\downarrow} \equiv f^s_{k,k'} + f^a_{k,k'}$, and $f_{k\uparrow,k'\downarrow} = f_{k\downarrow,k'\uparrow} \equiv f^s_{k,k'} - f^a_{k,k'}$. Also, $F^{s/a}_{k,k'} = N(\varepsilon_F) f^{s/a}_{k,k'}$; derive the Fermi liquid correction to the Pauli spin susceptibility χ in terms of $\chi^{(0)}$. You can assume an isotropic Fermi surface (i.e. it independent of the direction of $\vec{k}$), and can use, $N(\varepsilon_F)/N^{(0)}(\varepsilon_F) = m^*/m$.

[points: 3+7=10]